

Diagnostic report: Mg I

Computed levels: 5

Matched to NIST: 5

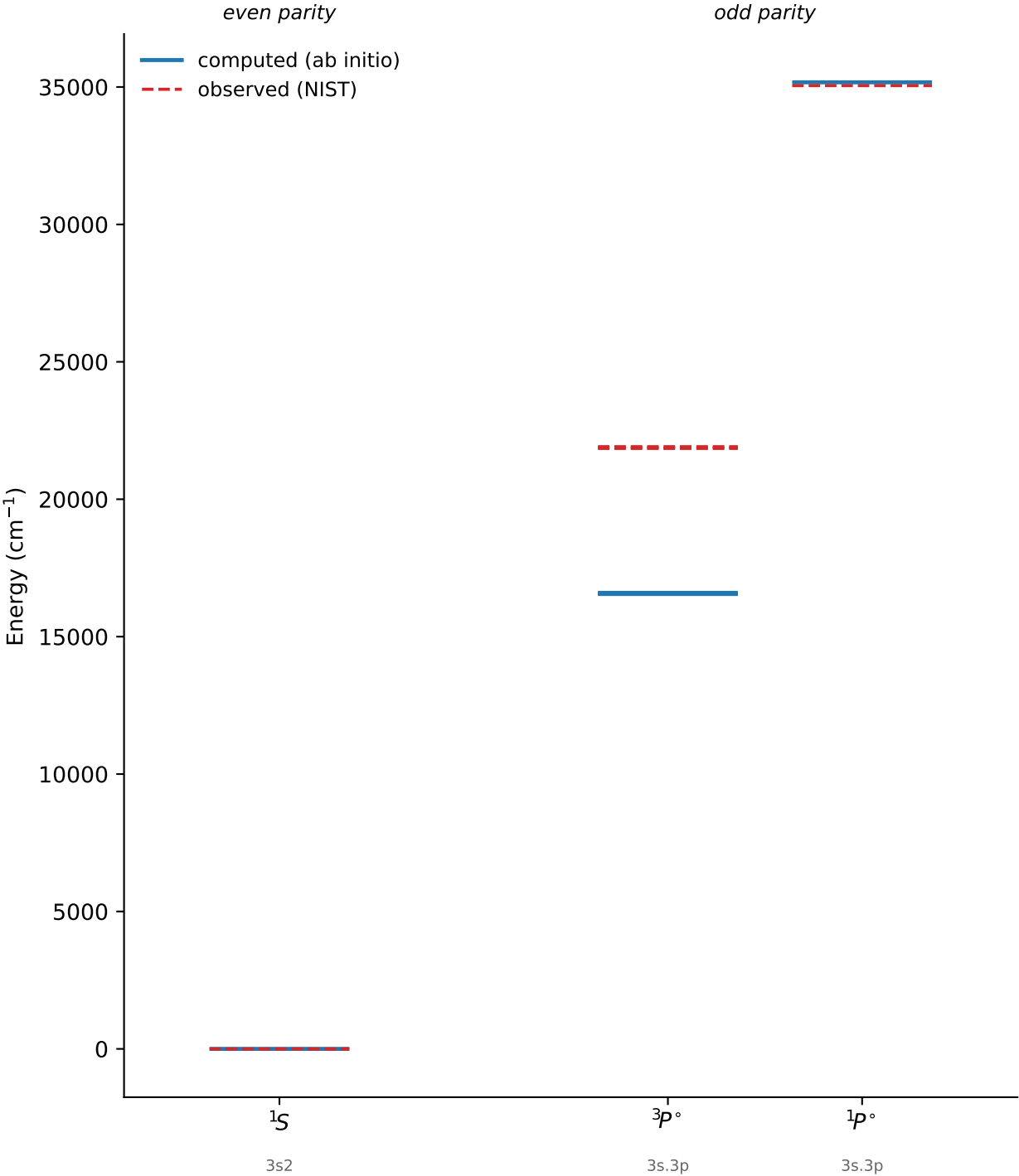
Computed E1 lines: 1

Pre-fit RMS ($E_{\text{calc}} - E_{\text{obs}}$): 4111 cm^{-1} (ab initio, no RCE fit yet)

Strongest computed lines:

$\lambda(\text{\AA})$	log gf	upper term
2843.34	+0.405	(2S) 1P

Mg I: term (Grotrian) diagram



Mg I: levels (computed vs observed)

config	term	J	E_calc	E_obs	ΔE (cm ⁻¹)
3s2	¹ S	0	0.0	0.0	+0.0
3s.3p	³ P°	0	16552.0	21850.4	-5298.4
3s.3p	³ P°	1	16566.0	21870.5	-5304.5
3s.3p	³ P°	2	16594.0	21911.2	-5317.2
3s.3p	¹ P°	1	35170.0	35051.3	+118.7

Mg I: level-fit residuals (RMS = 4111 cm⁻¹, ab initio)

